

Kiyan Amirian

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Research Summary

PhD student in Materials Science and Engineering at the University of Maryland, working at the intersection of machine learning and computational materials science. Research focuses on diagnosing where machine-learned interatomic potentials fail in realistic atomistic simulations, spanning force prediction, phase stability, elemental ordering, and ion migration, to guide the development of more reliable models.

Education

University of Maryland

Doctor of Philosophy in Materials Science and Engineering

Cumulative GPA: 4.0/4.0

College Park, MD, USA

2023-Expected Aug 2027

Shiraz University

Bachelor of Science in Materials Science and Engineering

Cumulative GPA: 17.15/20 (3.5/4.0); Last 2-year GPA: 18.54/20 (3.9/4.0)

Shiraz, Fars, Iran

2018-2023

Research Interests

Computational Materials Science

Atomistic Simulation

Machine-Learned Interatomic Potentials

AI for Science

Publications

K. Amirian, R. S. Saravanan, F. Adams, C. E. Schwarz, Y. Mo

arXiv, 2026

FP-DeErr: Application-Oriented Error Decomposition for Foundation Potentials

[arXiv:2609.05714](#) · [GitHub](#) · [Leaderboard](#)

K. Amirian, Z. Abbasi, R. Ebrahimi

2023

Analytical and numerical instability analysis of functionally graded low-carbon steel. International Journal of Iron and Steel Society of Iran, 19(2), 66–72. [doi:10.22034/IJISSI.2023.1990580.1262](https://doi.org/10.22034/IJISSI.2023.1990580.1262)

Research Experience

Graduate Researcher, UMD (Advisor: Prof. Yifei Mo)

2024-Present

- Developed FP-DeErr, an open-source, application-oriented error-decomposition framework for foundation potentials (FPs) that resolves errors by the physical quantities, configurations, and workflow stages governing each computational task; built its [evaluation code](#) and [public leaderboard](#), benchmarking 18+ FPs against DFT (ASE, MatCalc, pymatgen).
- **Force prediction:** decomposed force errors into magnitude and angle, and designed metrics for highly accurate predictions, large-force-error atoms, and far-from-equilibrium atoms (MatPES, OMat24), showing that these failure modes are diluted in average MAE/RMSE ([GitHub](#), [leaderboard](#)).
- **Phase stability and elemental ordering:** designed convex-hull agreement metrics (ground state, within-phase and global hull minimum) and evaluated ordering rankings (Top-1, Recall@k, Spearman's ρ) on chalcogenide phase-change materials, separating intrinsic energy-ranking errors from FP-relaxation errors ([GitHub](#), [leaderboard](#)).
- **Ion migration:** benchmarked FP-NEB against DFT-NEB on 154 Li- and Na-ion pathways with metrics for convergence, profile shape, endpoint energy ranking, and barriers, tracing barrier errors to intrinsic PES error, endpoint relaxation, and along-path forces ([GitHub](#), [leaderboard](#)).

Undergraduate Researcher, Shiraz University (Advisor: Prof. Ramin Ebrahimi)

2019-2022

- Developed an algorithm to predict tensile instability in functionally graded low-carbon steel and compared its predictions with published experimental data ([paper](#), [code](#)).

Selected Projects

<i>KV Cache Optimization for Large Language Models</i> Led a team comparing KV-cache compression techniques (StreamingLLM, KIVI, ZipCache) for long-context LLM inference. Reimplemented KIVI 2-bit quantization for LLaMA-2 and designed evaluation workflows across summarization, reasoning, and QA benchmarks, adding metrics for token match rate, entity hallucination, and sentence compliance (GitHub).	<i>Fall 2025</i>
<i>Image Classification: Classical ML and Transfer Learning</i> Built pipelines for MNIST and a 10-class species dataset using logistic regression from scratch, RBF SVM+PCA on MNIST (97.1%), and fine-tuned VGG19 transfer learning (97.06%) (GitHub).	<i>Spring 2025</i>
<i>Face Recognition: Dimensionality Reduction and Kernel Classifiers</i> Benchmarked PCA/MDA with Bayes, k-NN, SVM, and AdaBoost on a 200-subject dataset. Implemented a custom gradient-descent kernel SVM to address CVXOPT numerical failures with high-degree polynomial kernels (GitHub).	<i>Spring 2025</i>
<i>Electron-Beam Powder Bed Fusion Thermal Simulation</i> Wrote a FORTRAN DFLUX subroutine in Abaqus for a moving Gaussian electron-beam heat source on a Ti-6Al-4V powder bed, with temperature-dependent properties (GitHub).	<i>Spring 2022</i>

Skills

Programming: Python (NumPy, Matplotlib), Bash, MATLAB, Fortran, Git, $\LaTeX$
Atomistic Simulation: DFT (VASP), LAMMPS, ASE, pymatgen, MatCalc; NEB, molecular dynamics
Foundation Potentials (MLIPs): MACE, CHGNet, M3GNet, UMA, Orb, MatterSim, DPA4, eSEN
ML Frameworks: PyTorch, TensorFlow/Keras, scikit-learn
ML Methods: Transfer learning, CNNs, SVMs, LLM KV-cache quantization
AI Tools: LLM coding agents (Claude, ChatGPT) for developing, debugging, and validating research code
HPC: SLURM job arrays on university HPC clusters
Continuum Modeling: Abaqus (FEA)

Honors & Awards

Department of Materials Science and Engineering, University of Maryland <i>University of Maryland Dean's Fellowship Award, \$43,000</i>	College Park, MD <i>Spring 2023</i>
Department of Materials Science and Technology, Shiraz University <i>Ranked 2nd among 56 students in B.Sc. Materials Science and Engineering</i>	Shiraz, Fars, Iran <i>Spring 2022</i>

Relevant Graduate Courses

Introduction to Atomistic Modeling (4.0/4.0)	Natural Language Processing (4.0/4.0)
Statistical Pattern Recognition (4.0/4.0)	Machine Learning for Materials Science (4.0/4.0)

Teaching Experience

Kinetics, Diffusion and Phase Transformations (Fall 2024)	Mechanical Properties II (Spring 2022)
Fundamentals of Computer (Spring 2020)	Algebra (Fall 2019)

References

Yifei Mo Professor, MSE, UMD yfmo@umd.edu	Oded Rabin Assoc. Professor, ECE/MSE, UMD oded@umd.edu	Haizhao Yang Affiliate Assoc. Prof., Math, UMD hzyang@umd.edu
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